

India's Flex-Fuel Push: Navigating Between Policy Momentum and Consumer Economics



The Indian automotive industry is increasingly embracing multi-powertrain strategies, software-defined vehicles (SDVs), and, of course, the biggest attention-grabber - electric vehicles. Amid this transition, flex-fuel vehicles are emerging as a serious industry talking point and are steadily moving from policy discussion to early-stage industry adoption.

The government has been proactive in accelerating the ethanol-blending fuel roadmap and the reasons are not hard to understand. India spends Rs 22-23 lakh crore annually on crude oil imports, making the reduction of dependence on imported fossil fuels and the strengthening of energy security a strategic priority. India's ethanol production capacity is estimated to be around 2,000 crore litres, but blending demand is still developing with the nationwide transition toward E20. This structural gap has pushed policymakers to further encourage ethanol blending. On the flex-fuel infrastructure front, India is still in the early stages, but the government has launched the first E85 dispensing station in New Delhi and lined up plans to scale up to 500 stations by 2026 and around 5,000 by 2027-end.

The automotive industry has also responded to the government's ethanol mobility push with the country's largest carmaker Maruti Suzuki unveiling India's first flex-fuel car, the WagonR, which is

E20 to E100-complaint. India's largest two-wheeler maker joined the bandwagon, rolling out new E85-compliant flex fuel motorcycles - Splendor Plus and HF Deluxe (both comes with an E85-compatible engines). These vehicle launches are a good augury and will help built momentum around flex-fuel vehicles, but the critical question remains: what is the real consumer value proposition? Consumers must see a compelling reason to switch to flex fuel vehicles from their regular petrol vehicles (standard E20 petrol fuel) as adoption will ultimately boil down to economics.

The pricing of flex-fuel vehicles will be the key. Flex-fuel vehicles will need to compete directly with regular petrol vehicles using the baseline E20 fuel. It will be interesting to see whether the government will offer GST incentives (lower GST duty) to boost adoption of such vehicles. The fuel price of flex fuel vehicles is equally important. E85 is being offered at a lower price (around Rs 20 cheaper than regular E20 fuel), but whether this price difference would be adequate enough to drive mass adoption remains to be seen.

Ethanol-blended vehicles are generally less fuel-efficient than petrol vehicles. These vehicles can lead to a 25-30% drop in fuel efficiency, especially at higher blends such as E85 or E100. Although ethanol fuel is cheaper than regular E20 petrol fuel, the savings may not fully offset lower mileage and higher upfront vehicle costs, creating a clear trade-off between fuel price and efficiency.

There is a line of thought over how older vehicles using the regular E20 base petrol fuel will handle higher ethanol blends. Would these older vehicles be compatible with E85 or E100, in terms of performance and durability. It is likely that many of these vehicles will continue using E20, leading to a scenario system where E20 remains the dominant mainstream fuel while higher blends serve a narrower flex-fuel segment.

Globally, ethanol-based flex-fuel vehicles have been around for decades in Brazil. The Brazilian government built a robust ethanol ecosystem following the 1973 oil crisis, to reduce dependence on imported oil. Today, E27 (27% ethanol + 73% gasoline) is the standard fuel, and flex-fuel vehicles in Brazil can run on E100 (typically 93–96% ethanol with a small amount of water). This creates a fully flexible fuel ecosystem supported by strong infrastructure and consumer choice.

The United States has largely followed a blending-first model, through the Renewable Fuel Standard (RFS) that was introduced in 2005 and subsequently expanded in 2007. E10 is the standard fuel in the US, while E15 is gradually expanding - E85 is available mainly for flex-fuel vehicles. Ethanol in the US is primarily corn-based and used as a blending component, not as a full replacement fuel.

Other countries are also upping the ethanol-blending game. Thailand has developed a focussed ethanol-blend market - it offers 3 ethanol blends - E10, E20, and E85, with E20 strongly promoted by the government for regular passenger vehicles. Ethanol-blending fuel adoption is high in Colombia where E10 is the standard fuel. Although ethanol fuel vehicle penetration is smaller compared to Brazil, they have built a well-structured policy-driven model for gasoline substitution in Latin America. The Philippines has maintained an E10 blending mandate since the late 2000s, driven by energy security objectives. In Europe, Sweden was an early adopter of E85 flex-fuel vehicles in the 2000s, but they have increasingly shifted its transport strategy toward electrification, reducing ethanol's role over time.

Flex-fuel vehicles in India represent a policy-driven push to reduce fossil fuel imports and diversify energy sources. But their long-term success will depend less on policy intent and more on whether they make economic sense for consumers. While global markets like Brazil demonstrate that ethanol mobility can scale with strong infrastructure, India is still in the early stages of balancing infrastructure, pricing, and vehicle adoption. The key question is whether lower fuel costs can outweigh lower fuel efficiency and higher upfront vehicle costs. If aligned effectively, flex fuels could emerge as a meaningful transitional pathway alongside electrification.